

SENTINEL AI Fraud Detection

Transaction Risk Analysis Report
Generated: 2026-10-07 16:09 UTC

Risk Level: SUSPICIOUS — Score: 34.4/100

Transaction Details

Transaction ID	TXN0000360089
Customer ID	CUST00049761
Date / Time	2024-12-30 23:58:00
Amount	\$4.03
City	Mexico City
Country	USA
Merchant Category	ATM Withdrawal
Payment Method	Mobile Payment
Device Type	POS Terminal
International	No
Night Transaction	Yes
Distance from Home	56.6 km
Time Since Last Txn	13.07 hours
Failed Attempts	2
PIN Changed Recently	No
Actual Fraud Label	NORMAL
Fraud Type	Unknown

Customer Attributes

Customer Age	63
Credit Score	721
Account Age (years)	3.4
Account Balance	\$6,122.27
Prior Transactions	155
Monthly Frequency	16

Model Prediction

Prediction	NORMAL
Risk Level	SUSPICIOUS
Risk Score	34.42 / 100
Anomaly Score	0.2801
Reconstruction Error	0.332217

Risk Factor Breakdown

Factor	Score	Weight	Contribution	Explanation
Geographic Risk	0.800	0.15	12.00	Very long-distance transaction: 9847 km from Munich to Mexico City (13.1 hours since last transaction).
Autoencoder	0.280	0.35	9.80	Transaction pattern appears relatively normal (autoencoder score: 0.28).
Time Anomaly	0.700	0.10	7.00	Transaction at 23:00 — unusual off-hours activity (night transaction flag set).
Velocity Risk	0.271	0.10	2.71	Transaction 13.1 h after last — within normal velocity range.
Behavioral Deviation	0.188	0.08	1.50	Behavioural deviations detected: unusual city 'Mexico City', unusual payment method 'Mobile Payment', unusual device 'PO
Failed Attempts	0.667	0.02	1.33	2 prior failed attempts — moderate credential-stuffing risk.
Credit Factor	0.064	0.01	0.06	Good credit score (721) — low credit risk.
Amount Deviation	0.000	0.15	0.00	Transaction amount \$4.03 is 0.8 σ below customer's average \$114.27 (σ =\$135.20).
International Flag	0.000	0.03	0.00	Domestic transaction — no international risk.
Pin Change	0.000	0.01	0.00	No recent PIN change.

Customer Behavioural Profile

Average Amount	\$114.27
Median Amount	\$75.76
Std Dev Amount	\$135.20
Common City	Munich
Common Country	USA
Common Category	ATM Withdrawal
Common Payment	Credit Card
Common Device	Desktop
Avg Transaction Hour	13.8:00
Txns / Day	0.00

Total Transactions	0
Fraud Count	0